

Role of Software Vendors in Technology Adoption in the AEC Projects' Context: A Comparative Review

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ABSTRACT:

The AEC industry is experiencing a significant technological thrust driving the innovation of Building Information Management (BIM), Digital Twins, Digital-Fabrication, Robotics, Modularization, Virtual Collaboration, and Integrated Design and Construction workflows while leveraging Artificial Intelligence, Lean principles, and Sustainability practices. One of the significant barriers to technology adoption in the AEC domain is the lack of sufficient education and training, i.e., appropriate knowledge, skills, and abilities (KSAs) that allow combining the technical design and construction know-how with the technological expertise to produce accurate results effectively. Along these lines, in addition to the major stakeholders of the AEC industry, software and hardware vendors have a crucial role in driving technology adoption. While this is true, challenges exist in how such vendors knowledgeably support technology adoption that is effective and sustainable. Addressing some of these challenges, this study focused on elaborating on the role of software vendors in technology adoption in the context of AEC projects. A comparative study was conducted by reviewing various academic and industry publications and selected software vendor websites. It was noted that software supports the AEC project delivery process in many ways. Some are generic, supporting generic project functions such as scheduling and estimating; some can be customized to the user to manage a specific process in the organization, such as change order management; some are combinations of both. The vendor's business model and hiring practices are often structured based on these distinctions. This paper presents the summary of this study, providing a state-of-the-art report on this topic completed by a master's student in the Construction Management Program at California State University-Fresno.

Keywords: Digital project delivery, BIM, Cloud collaboration, Artificial intelligence, software Vendors in AEC

INTRODUCTION:

The Architecture, Engineering, and Construction (AEC) industry is valued at over \$2.84 trillion. Innovations like Building Information Management (BIM), Digital Twins, and robotics are

driving a significant shift. (Agenda, 2016). Over the past decade, companies have increasingly turned to digital solutions to streamline construction due to labor shortages and a demand for skilled workers (Gledson et al., 2024). Software vendors need to integrate new technologies to improve construction operations and management. The construction technology industry has remained resilient, with a steady investment of around \$5.38 billion for 2022 (“CEMEX Ventures - Fostering the construction revolution,” n.d.; “Digital Construction Software | Autodesk,” n.d.). This study explores how digital technologies enhance project delivery and improve productivity, safety, and quality through stakeholder collaboration. The primary objectives are as follows:

1. Examine the AEC Technology Landscape. Investigate current technological trends in the AEC industry, focusing on innovations like BIM, digital twins, robotics, and AI.
2. Assess the contributions of software and hardware vendors in overcoming technology adoption barriers by analyzing their support mechanisms and challenges.

SUMMARY:

The lack of education and training is a significant barrier to technology adoption in the AEC industry. Software and hardware vendors must drive technology adoption in this sector. (Zhang et al., 2023). A study was conducted to address the challenges in technology adoption within AEC projects and explore software vendors' role in overcoming these challenges. The research involved a qualitative analysis, which included a literature review and a review of vendor websites to identify how companies provide education and training to bridge the gaps. The study also compared how software supports various aspects of the AEC project delivery process. This paper offers a condensed overview of the research, providing valuable insights into the dynamic landscape of the AEC industry and contributing to the discourse on technology integration in AEC projects.

LITERATURE REVIEW :

The literature review focuses on the current state of Digital Project Delivery and how different technologies are being utilized in the design and construction industry. We will explore the importance of collaboration platforms, the involvement of vendors, and a variety of existing, emerging, and future technologies that are influencing the sector.

1. Digital Project Delivery : An Introduction

Digital Project Delivery refers to a new way of managing construction projects that has been defined by the National Institute of Building Sciences and Autodesk. It involves the use of cloud-based collaboration platforms and advanced digital technologies to allow better communication and coordination among project stakeholders (Autodesk et al., 2024). Digital Project Delivery helps improve project outcomes by increasing efficiency, accuracy, and cost-effectiveness. It simplifies workflows, optimizes resource utilization, and reduces risks, promoting innovation and progress in the construction industry.

2. Vendors Contributions in Design and Construction Technologies and Website Review:

Construction companies use digital technologies and cloud-based collaborative platforms to overcome project challenges. Digital Lean principles such as Streamline, Digital pull planning,

Visual management, LPS, Takt planning, and Kanban are being integrated to enable stakeholders to provide feedback, estimate, visualize, establish quality and safety standards, and make financial decisions within a unified platform. Vendors like Procore, Nialli, smartapp.com, and Bulldots offer project management software, Lean Construction tools, and digital planning solutions to revolutionize the construction industry(Dallasega et al. 2020). Lean Construction Institute (LCI) and companies like Smartapp, Velisun, V-planner, Clear Flow Matrix, Takt Planning, Autodesk, and safety platforms are driving the industry's digital transformation by offering digital planning tools, BIM and design coordination, clash detection, estimation, and digitized safety procedures. This enhances collaboration, efficiency, safety, and cost-effectiveness throughout the construction lifecycle.

3. Existing Design and Construction Technologies :

The literature review analyzes recent design and construction technologies, highlighting their importance in addressing industry challenges. AutoCAD is popular software for creating 2D plans, 3D designs, and structural analyses.(Onstott 2014; Palm 2020). BIM platforms like Revit, SketchUp, Tekla, and Navisworks enable detailed design and analysis with multiple views, collaboration, sequencing, and scheduling.(Chen and Luo 2014; Mesároš et al. 2016). Yulio Woodhouse offers VR solutions for client communication, design validation, estimation, inspection, and safety training. IoT technology, used by companies like Giotec Science and DeWalt, enhances worksite safety and productivity.(Balzan et al. 2020; Ghosh et al. 2021). Innovative robotics, such as Hadrian X, and 3D printing robots have revolutionized construction processes. AI systems, such as Daisy, Krates, Plannerly, and Autodesk, optimize design, automate tasks, and enhance safety. These technologies are driving construction's digital transformation, improving collaboration, efficiency, safety, and cost-effectiveness.

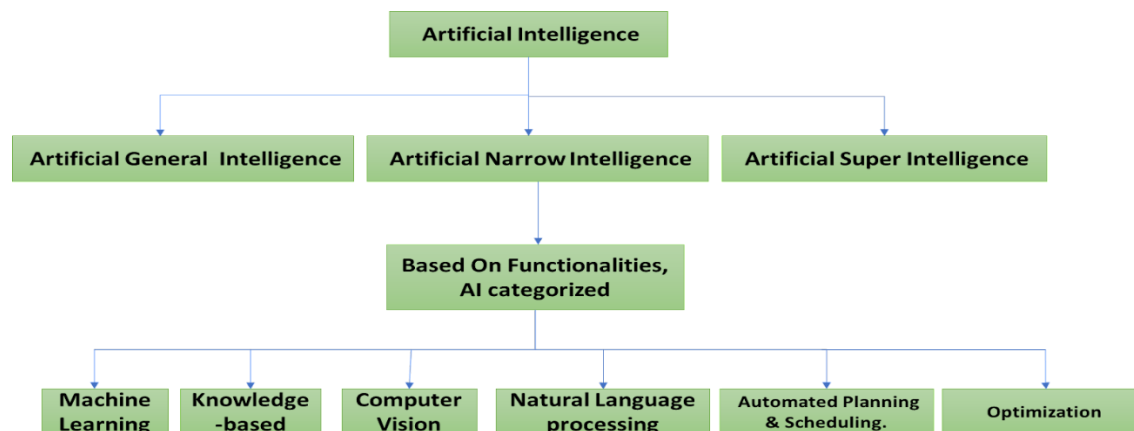


Figure 1. Types of Artificial Intelligence

4. Lean Methods Alignment of Technologies:

The construction industry has witnessed a proliferation of software solutions aimed at optimizing project management while adhering to Lean principles. Smartapp.com provides a cloud-based platform integrating planning, management, safety, and dispatch, enhancing collaboration (Smartapp). Nialli employs a Pull planning system and Visual planning to foster collaboration and constraint identification (Nialli). In Takt streamlines project planning via a web-based cloud solution, improving methodologies (In Takt). VeilSun emphasizes Lean implementation across

Planning, Construction, and Safety through digital Pull planning (VeilSun). Clear Flow Matrix supports Lean principles with simplified planning methodologies (Clear Flow Matrix). V Planner promotes Lean thinking and collaboration (V Planner). Buildots reduces rework and enhances efficiency (Buildots). Hoylu for Construction promotes Lean methodologies through visual planning (Hoylu). PM Web offers lifecycle management with visual management (PM Web). These solutions collectively drive Lean implementation by enhancing collaboration, streamlining processes, and optimizing project management practices across phases.

METHODOLOGY:

To conduct this research, a qualitative analysis was performed. The analysis utilized select research papers that were obtained through keyword searches. The searches included terms such as BIM, digital technologies, AI, vendors in AEC, and digital lean project delivery. These searches were conducted on a variety of platforms, such as Google Scholar, IEE, Engineering Village, Research Gate, and vendors' company websites. In total, 158 publications were collected. Relevant information was recorded on a Google sheet, including details such as the paper's title, citations, abstract, keywords, methodology, lean alignment, future recommendations, and more. Additionally, a thorough review of vendors' websites was conducted. The information gathered was recorded in the same Google sheet. This included information such as the company name, technology focus, primary focus, summary, and lean implementation. The information was carefully collected in a systematic manner, utilizing keywords, abstracts, and phrases such as BIM, Lean, digital project delivery, and how companies are leveraging technology in AEC projects. To evaluate all the information gathered by this qualitative analysis, research questions, organization questions, and conclusion questions were prepared. These will be discussed with the vendors during the conference.

RESULTS AND DISCUSSION:

DPD is revolutionizing AEC by using cloud-based platforms and digital tools to enhance collaboration, communication, and coordination among stakeholders. Experts define DPD as a new era in construction project management. Software vendors like Procore, Nialli, and Bulldots provide project management software, Lean Construction tools, and digital planning solutions to streamline workflows and mitigate risks (Autodesk et al., 2024). Modern construction technologies like BIM, Navisworks, and IoT have significantly improved worksite safety, productivity, and project efficiency. Yulio Woodhouse provides VR solutions for communication improvement, and Smartapp.com, Nialli, and VeilSun promote Lean principles for construction planning. These technologies foster innovation and lead to a more sustainable built environment.

CONCLUSION :

The AEC sector is being transformed by Digital Project Delivery (DPD) and software vendors. DPD involves cloud-based collaboration platforms and advanced digital tools, improving communication, coordination, and decision-making among project stakeholders. Software vendors offer tailored solutions that streamline workflows, optimize resource utilization, and mitigate project risks. Existing and emerging design and construction technologies like BIM, Revit, Navisworks, VR, IoT, robotics, 3D printing, and AI enhance communication, safety,

productivity, and efficiency on construction sites. These technologies drive efficiency, innovation, and sustainability in construction project delivery, shaping the future of the AEC industry. Embracing these technologies and fostering strategic partnerships with software vendors will be essential for achieving success in construction projects.

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